

Modeling the Spectral Energy Distributions of Binary Supermassive Black Holes

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Abstract

This Project presents a computational graphing of Spectral Energy Distributions (SED) for Binary Supermassive Black Holes (SMBH). This project consists of couple main parts. We start with Exploring Blackbody radiation equation, then we model single quasar accretion disk, then we adapt the single quasar SED to model a circumbinary disk and then finally we add in two mini disks and model a whole Binary system. The result is that a binary SMBH SED has a different quality then a Single quasar which is a cavity in between of the mini disks and the outer circumbinary disk. This shape is different than a single Quasar which might provide pathway to identify binary SMBH candidates observationally.

Introduction

Super Massive Black holes, with masses ranging from millions to billions of our Sun are found in centers of galaxies. When two galaxies merge, the SMBHs in their centers are expected to form a gravitational bond which ends up in a Binary SMBH system. These systems are difficult to observe because they have similar characteristics to Active Galactic Nuclei (AGN) or a Quasar. One promising pathway is SED observation. The distribution of emission as a function of wavelength, A binary SMBH system is expected to have a qualitatively different SED, by nature it should have a central cavity in the circumbinary disk and by additional emission from smaller mini disks around each black hole.

The goal of this project is to graph and compare these SEDs. Building from a single Quasar to a Binary SMBH system.

Background and Theory (Equations included)

Blackbody Radiation and Planck's Law

A blackbody is an idealized object that absorbs all incident radiation and re-emits it based solely on its temperature. The spectral radiance of a blackbody, the power emitted per unit area per unit wavelength is given by Planck's Law:

$$B(\lambda, T) = (2hc^2/\lambda^5) \times 1 / [\exp(hc / \lambda kT) - 1]$$

Code Implementation:

```
def plancksformula(wavelength, T):  
    exponent = (h * c) / (wavelength * k * T)  
    numerator = 2 * h * c**2  
    denominator = (wavelength**5) * (np.exp(exponent) - 1)  
    return numerator / denominator
```

Thin Accretion Disk Theory

An accretion disk forms when matter falls toward a compact object with angular momentum, causing it to spiral inward in a flattened rotating structure. For a geometrically thin, optically thick disk, Shakura and Sunyaev (1973) showed that the disk radiates locally as a blackbody at each radius, with a temperature that depends on the mass of the central object, the accretion rate, and the radius.

$$T(r) = T_0 \times (r/r_{in})^{(-3/4)} \times [1 - \text{sqrt}(r_{in}/r)]^{(1/4)}$$

where r_{in} is the inner radius of the disk (typically the innermost stable circular orbit, ISCO), and T_0 is a normalization temperature that depends on the black hole mass M , the accretion rate $\dot{M}$, and the Stefan Boltzmann constant:

$$T_0 = [3GM\dot{M} / (8\pi \sigma r_{in}^3)]^{(1/4)}$$

This profile captures the fact that the disk is hottest near the inner edge, and cools as a power law with increasing radius. The inner factor accounts for the zero-torque boundary condition at the ISCO. In early stages of this project, the simplified version $T(r) \propto r^{(-3/4)}$ was used, dropping all constants, as an first approximation.

Code Implementation:

```
def temperature_profile(r, T0, r_in):
    term = 1.0 - np.sqrt(r_in / r)
    if term <= 0:
        return 0.0
    return T0 * (r / r_in) ** (-3/4) * term ** 0.25

def T0_shakura(M, Mdot, r_in):
    return (3 * G * M * Mdot / (8 * np.pi * sigma * r_in**3)) ** 0.25
```

Eddington Accretion Rate

The Eddington luminosity is the maximum luminosity at which radiation pressure balances gravity for a spherically accreting object:

$$L_{Edd} = 1.26 \times 10^{31} \times (M / M_{sun}) \quad [Watts]$$

The corresponding Eddington accretion rate is:

$$\dot{M}_{Edd} = L_{Edd} / (0.1 \times c^2)$$

Code Implementation:

```
def eddington_mdot(M):
    L_edd = 1.26e31 * (M / M_sun)
    eta = 0.1
    return L_edd / (eta * c**2)
```

Binary SMBH Disk Structure

In a binary SMBH system, the disk is made up of three main parts: one outer circumbinary disk and two smaller mini disks around each black hole. The circumbinary disk surrounds both black holes, while each black hole has its own mini disk inside the central gap. This is the most basic structure of a binary SMBH disk system and this is what is assumed in this simulations.

Figure 1:

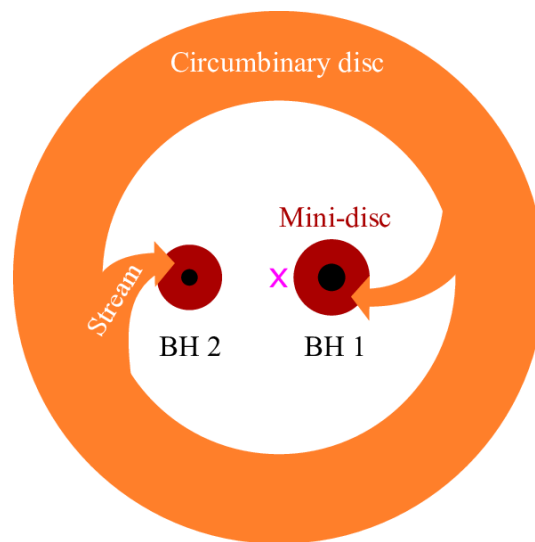


Figure taken from Liao et al. (2023).

Results

The full binary SMBH SED is shown in Figure 2, divided into its three physical components: the circumbinary disk, and two mini disks surrounding each black hole. This graph shows the normalized emission of each component, making it easy to compare their spectral shapes independently of their absolute luminosities. The mini disks, with peak temperatures of 1.8×10^5 K and 2.2×10^5 K, emit most of their energy at shorter wavelengths, while the cooler circumbinary disk $T_0 \sim 1.5 \times 10^4$ K peaks at longer optical wavelengths $\sim 0.5 \mu\text{m}$. This separation in peak wavelength is a direct consequence of the temperature between the components.

Figure 2:

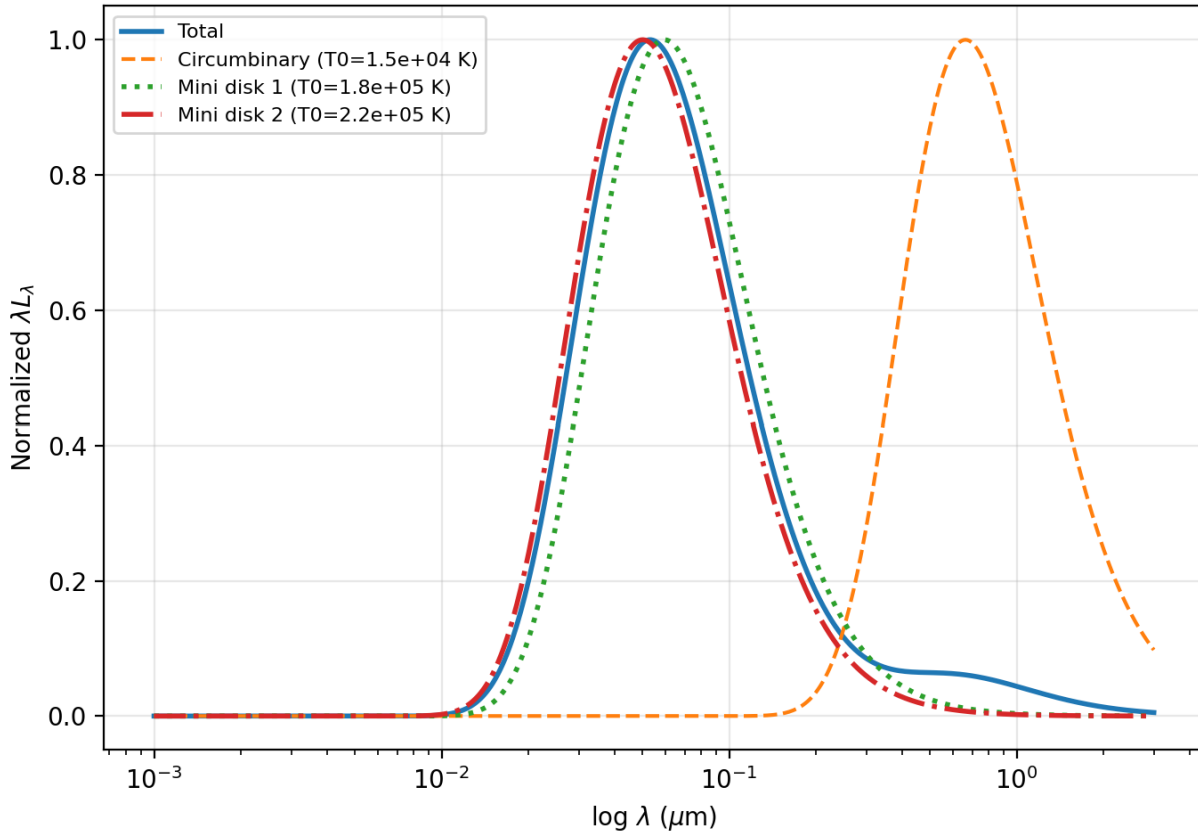


Figure 3 shows the physical SED of the binary SMBH system broken down into its three individual components on a log-log scale. The two mini disks (green and red) are compact and hot, peaking at short wavelengths around $\sim 0.01\text{--}0.05\ \mu\text{m}$ and dominating the total emission at those wavelengths. The circumbinary disk (orange) is much larger and cooler, and as a result its emission peaks at much longer optical wavelengths around $\sim 0.5\text{--}1\ \mu\text{m}$. The most notable feature in the total SED (blue) is the drop that occurs between roughly $0.05\text{--}0.5\ \mu\text{m}$ this is the region between where the mini disk emission falls off and where the circumbinary disk emission picks up. This gap corresponds directly to the central cavity in the binary system, the region cleared of gas by the gravitational torques of the two orbiting black holes. It is this drop that makes the binary SED physically distinct from a single quasar, where warm intermediate radius disk material would fill in this region and produce smooth continuous emission with no such deficit

Figure 3:

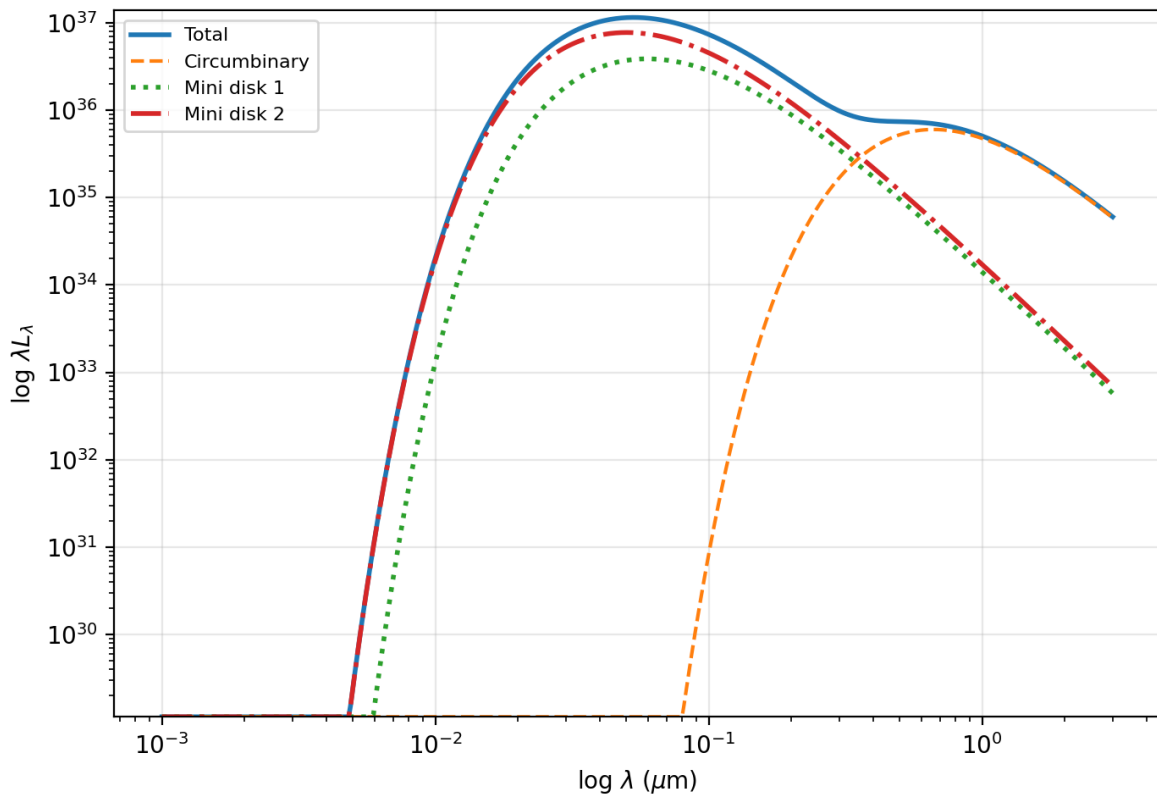
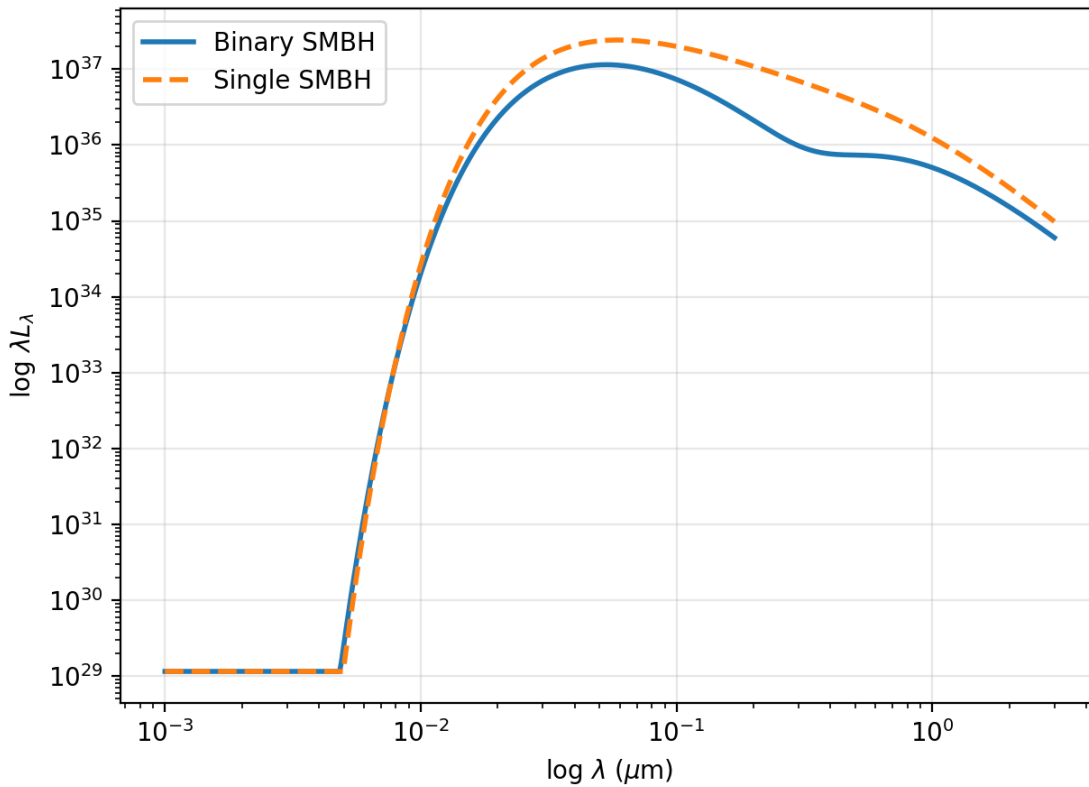


Figure 4 presents the key result of this project: a direct comparison between the total binary SMBH SED (blue) and a single quasar SED (orange dashed) of the same total mass and accretion rate. At the shortest wavelengths both curves are nearly identical, as the hot mini disks in the binary system produce UV emission comparable to the hot inner regions of the single quasar disk. However, the binary SED drops clearly below the single quasar, producing a spectral deficit before partially recovering at longer wavelengths where the circumbinary disk begins to contribute. The single quasar shows no such feature its declines smoothly and continuously across the entire wavelength range, exactly what you would expect from a disk with no central cavity. This divergence between the two curves is the most important result of this project, it can be used as a identifier for Binary SMBH systems.

Figure 4:



Discussion

The results show that a binary SMBH system produces a measurably different SED compared to a single quasar of the same mass. The key difference is the spectral deficit (the dip), which arises directly from the central cavity. In a single quasar this region is filled by warm continuous disk emission, so the absence of that emission in the binary case is a physical signature of the binary structure. The mini disks partially compensate at short wavelengths, which is why the two SEDs look similar there, but the deficit in between is hard to hide. This suggests that observation of SEDs can be used to identify binary SMBH candidates.

That said, this model has limitations. The disk radii and accretion rates are assumed rather than derived self-consistently. Gas streams flowing through the cavity could also add emissions that partially fills in the deficit, making the signature harder to detect in practice. These are all things that would need to be addressed in a more complete model.

Conclusion

This project built a computational model of binary SMBH SEDs from first principles, starting with Planck's blackbody law and arriving at a three component binary model, circumbinary disk plus two mini disks with parameters drawn from Shakura-Sunyaev thin disk theory. The final comparison between single and binary SMBH SEDs shows a clear spectral deficit at intermediate UV wavelengths and a secondary optical shoulder in the binary case, both of which are direct consequences of the central cavity carved by the binary's gravitational torques. These features are physically meaningful and potentially observable, suggesting that broadband SED shape could serve as a diagnostic for binary SMBH candidates

References

Shakura, N. I., & Sunyaev, R. A. (1973). Black holes in binary systems. Observational appearance. *Astronomy and Astrophysics*.

Liao, S., Johansson, P. H., Mannerkoski, M., Irodotou, D., Rizzuto, F. P., McAlpine, S., Rantala, A., Rawlings, A., & Sawala, T. (2023). Modelling the accretion and feedback of supermassive black hole binaries in gas-rich galaxy mergers. *MNRAS*.